

Solution Requirements

Date	07 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	System operates as a secure localized workspace session. Future scopes include setting up light API token authentication validation keys.	Low
2	Authorization levels	Default tier provides comprehensive read/write execution accesses for standard local single-user profiles to manipulate individual metrics.	Medium
3	External interfaces	The platform must establish a stable HTTPS socket handshake out to the public Hugging Face Hub (model downloads) and the server Wikipedia API text search wrapper .	High

4	Transactions processing	Core system processing manages local JSON standard updates, ensuring concurrent multi-tab frontend queries do not lock or corrupt file buffers during execution loops.	High
5	Reporting	The web interface must expose a responsive metrics section calculating exact user click counts (Thumbs Up vs. Thumbs Down) and computing live success ratios.	High
6	Business rules, etc.	The web interface must expose a responsive metrics section calculating exact user click counts (Thumbs Up vs. Thumbs Down) and computing live success ratios.	High
7	Compliance to laws or regulations	The script runs localized memory parsing systems. It avoids extracting user personal data assets, preserving privacy compliance standards automatically.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	FastAPI framework response latency for processing local inference predictions should minimize network delays.	API endpoints must return payload components in less than 2.5 seconds
2	Scalability	Python web server routing mechanisms must handle concurrent processing overhead efficiently without memory allocation crashes.	Supports clean concurrency handling for up to 50 concurrent local loop interactions
3	Security & Data Privacy	Local system file-handling layers should prevent unauthorized direct shell reading access to underlying file blocks.	Local path restrictions block random file folder navigation queries
4	Reliability & Availability	Error handling mechanisms inside model prediction handlers must catch missing or empty payload strings gracefully.	Application achieves a 99% fault-tolerant uptime rate during standard sandbox testing routines
5	Usability & Accessibility	Streamlit visual application states must remain fully readable, maintaining legible contrast layers for long testing durations.	UI layout transitions adjust layout viewports responsively across screen displays